DATA ENGINEERING · ORAL DEFENSE · NOVEMBER 2026

FreshRoute Logistics.

FreshRoute
LOGISTICS

"Fresh to the shelf, on time."

Are we keeping our delivery promise?

An end-to-end pipeline that turns four messy sources into delivery KPIs you can trust.

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Snowflake · dbt · Docker · Airflow · Terraform · Streamlit



WHY THIS PROJECT

A failed delivery is paid zero times, and driven twice.

Supermarkets pay FreshRoute **only per successful delivery**. A refused, damaged or “not home” parcel earns nothing and must go out again.

The contract asks for **85 % OTIF** (on time in full). But the answer to “where do we miss it?” is spread across four systems that never agree.

WHAT WE BUILT

01 **Messy sources**
CSV exports, Postgres, GPS stream, three public APIs

02 **One tested pipeline**
Bronze, silver, gold in Snowflake; dbt tests; daily Airflow run

03 **KPIs you can trust**
OTIF per depot, rain impact, lost fees in EUR

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1. The company.

A last-mile partner for UK supermarkets.

Supermarkets pay FreshRoute **per successful delivery** (1 to 10 £ depending on weight and slot).

A refused, damaged or “not home” parcel is **not billed** and must be delivered again.

STAKEHOLDERS

Depot managers, operations director, fleet manager, and a CFO who reports to a European parent group in EUR.

10
depots (UK + Paris)

~130
vans and refrigerated trucks

~15k
delivery records in September

85 %
contractual OTIF target



1. The problem.

Four systems, no shared truth.

Transport system

Depots, vans, routes · nightly CSV export

Ops database

Deliveries and outcomes · on-prem Postgres

Telematics

GPS pings from every van · JSON stream

Outside world

Weather, exchange rates, bank holidays

Which depot breaks its delivery promise?
Depot managers

Does rain make it worse?
Operations director

How much do failed attempts cost us, in EUR?
CFO, group reporting



2. Use cases.

Three decisions, three precisely defined metrics.

1 Depot manager

Moves drivers to failing depots

METRIC · GRAIN

OTIF = delivered before promised_by ÷ deliveries due · day × depot_id, quarantine excluded

ACCEPTANCE TEST

One depot-day recomputed in SQL = dashboard

2 Ops director

Adds slots on rainy days

METRIC · GRAIN

Success rate, avg late minutes by rain (dry, light, heavy > 10 mm) · depot × category

ACCEPTANCE TEST

Rows per category sum to fct_deliveries

3 CFO

Reports billed vs lost fees

METRIC · GRAIN

Billed = Σ fee delivered · Lost = Σ fee failed · EUR = GBP × weekly ECB rate · week × depot

ACCEPTANCE TEST

dbt test: gold total = silver total



3. Sources & ingestion.

Six sources, five ingestion methods.

TMS export

Depots, vans, routes · CSV

Batch: PUT → stage → COPY INTO

Nightly export, daily decision

Ops database

Deliveries · Postgres

Incremental: watermark + append

Only the delta moves

Telematics

GPS pings · JSON

Micro-batch stream, resumable

Simulates Kafka at no cost

Open-Meteo API

Weather archive

REST API → VARIANT

Real public data, no key

Frankfurter API

GBP → EUR (ECB rates)

REST API

Group reports in EUR

GOV.UK API

UK bank holidays

REST API

Explains activity dips



3. Architecture.

One image ingests, Snowflake stores, dbt refines.

- TMS CSV export
- Postgres ops DB
- GPS telematics
- 3 public APIs

→ **Docker**
Python ingestion jobs, retry, run id

→ **Snowflake · FRESHROUTE**

BRONZE	SILVER	GOLD
raw text + lineage	clean + quarantine	use-case tables

dbt build: models + tests, gold skipped on failure

→ **Streamlit**
ANALYST role, gold only

Airflow · daily at 05:00 London · retries, then Slack alert

Terraform · warehouse, database, schemas, roles, budget



3. Tool choices.

Each tool earns its place.

Snowflake

Storage and compute split, XS auto-suspend, idempotent COPY

vs BigQuery, pay per TB scanned

dbt Core 1.11

Versioned SQL, tests, lineage docs, cleaning macros

vs Dynamic Tables, no tests

Docker

One pinned image for every job and dbt, non-root

vs local venvs per laptop

Airflow

Dependencies, retries, alerts, replay of one day

vs Snowflake Tasks, lighter

Terraform

Warehouse, roles and budget as code, reviewable plan

vs clicks in Snowsight

Deliberately not used: Kafka, Debezium CDC

No decision is taken in real time: the depot briefing is at 08:00. CDC would only pay off if the ops database hard-deleted rows.



4. Silver layer.

Repair what is certain, quarantine what would lie.

Problem in the data	Example	Fix
Missing values as text	N/A, null, -, blank	clean_text macro → NULL
Mixed timestamps	Z, +01:00, epoch, dd/mm/yyyy, 2099	clean_ts → UTC, London business day
Messy money	£1,234.50 · €5.30 · 12,50 · pence	clean_money, ÷100 above 50 £
Category spelling	DELIVERED, failed_-_not_home	normalise + mapping seed
Broken JSON payload	“NULL”, LAT instead of lat	TRY_PARSE_JSON + COALESCE
Duplicates and updates	same id twice, day-2 update	QUALIFY: latest batch wins
Schema drift	outcome → outcome_v2	mapped, unknown column = warning
Would make a KPI wrong	unknown route, unreadable fee or date	Quarantine, with the reason

4. RECONCILIATION

Gold matches silver to the penny.

Billed fees in silver

£16,838.12

=

Billed fees in gold

£16,838.12

Keys and completeness
unique, not_null, every delivery has a known route

Business rules
fee between 0 and 50 £, quarantine under 15 %, gold = silver

Grain and freshness
one row per gold grain, dbt source freshness under 24 h

Source: Snowflake check query + dbt build (all tests pass)



5. Reliability.

Four incidents, four safe outcomes.

1 · API down

3 retries with backoff, then a loud failure and a Slack alert. dbt does not run.

2 · Same day replayed

COPY loads 0 files, watermark pulls 0 rows, APIs delete and insert. Totals unchanged.

3 · Corrupted day

4,000 deliveries with an unknown route: quarantine passes 15 %, the test fails, gold is skipped.

4 · Schema drift

Day 2 renames outcome to outcome_v2: the rename is mapped, unknown columns are logged.

5. Consumption.

The dashboard answers the depot question.



1. **Reads gold only**
   Read-only ANALYST role, cached 10 minutes

2. **£16,838 billed**
   The reconciled total from slide 12

3. **By depot_id, not city**
   Two depots in Paris, two in London

Source: Streamlit on FRESHROUTE.GOLD



6. Orchestration & IaC.

Scheduled by Airflow, declared in Terraform.

Airflow DAG · freshroute_daily

6 ingestion tasks → dbt_build → source freshness

- Every task runs in the same container image
- Scheduled 05:00 Europe/London, before the 08:00 briefing
- Batch comes from the run, so any day can be replayed
- 2 retries, then a Slack alert after the last one

Terraform · Snowflake provider

plan → review → apply → plan shows no drift

- Warehouse XS, auto-suspend 60 s
- Database with BRONZE, SILVER and GOLD schemas
- ENGINEER and ANALYST roles with grants
- Budget monitor: notify 75 %, suspend 100 %
- State never committed; remote S3 backend ready

Manual steps, stated: Snowflake account, state bucket, RSA public keys on the two service users.



6. Security & costs.

Least privilege, and a prototype under 10 $ a month.

Analyst sees gold only

Bronze query denied, gold query allowed (sql/checks.sql §6)

No passwords in the pipeline

Key-pair auth; key mounted at run time; .env and keys ignored by Git and Docker

Pseudonymised customers

Only a customer_ref, no name or address

Cost item	Estimate
Warehouse XS, ~6 min/day	~3 credits/month
In dollars	~6 to 9 $/month
Storage	under 1 GB, negligible
10× more depots	~30 credits/month, still XS

A budget alert is a warning, not a cap: the monitor’s suspend trigger is the cap. Cleanup: terraform destroy, docker compose down -v.

Assumptions: 1 credit per hour on XS, one run per day



7. Limitations.

What our numbers do, and do not, say.

1. **OTIF 12.3 % is not a finding**
   Generated outcomes are random, about 25 % each. We prove the KPI is computed correctly and flags every depot under target.

2. **No rain effect visible**
   Expected with random outcomes; the question becomes answerable with real data.

3. **Last week is partial**
   The dip at the end of the UC3 chart is an incomplete week, not a drop.

Next steps.

- Debezium CDC if the ops DB hard-deletes rows
- Snowflake ALERT with e-mail on quality checks
- Managed Airflow instead of a local scheduler
- Remote Terraform state in a locked bucket

Ask Questions to Nayé !



Danila · Nayé · Nicolas — thank you.

CODE AND README

github.com/ndollet-sudo/freshroute-data-engineering